

The equation of a straight line

Every straight line is one linear equation — and every linear equation is one straight line.

LESSON 37

1 × 40 MIN

GEOMETRY 8, TEMA 34

STAGE 9 · 10.1–10.2

LESSON CLOCK

5'

12'

12'

7'

4'

Plot three points and join them	5 min
Gradient and intercept	12 min
Through two points	12 min
Special lines	7 min
Homework	4 min

LEARNING OBJECTIVES

- Write the equation of a line from its gradient and a point.
- Find the equation of a line through two given points.
- Recognise the equations of horizontal and vertical lines.
- Decide whether a point lies on a line.

TEXTBOOK

National	Tema 34, pp. 75–77
Cambridge	Learner’s Book pp. 214–224
Workbook	Workbook 10.1

01

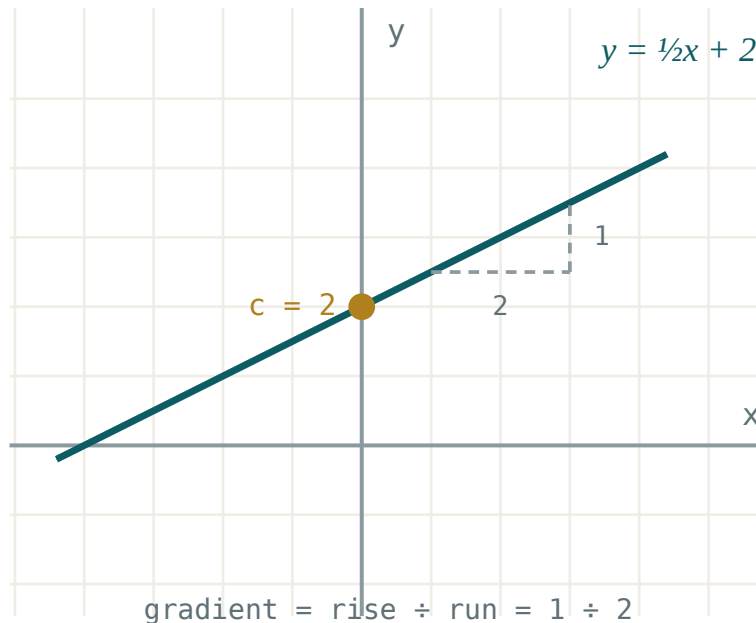
Explanation

The two standard forms

EQUATION OF A STRAIGHT LINE

$y = kx + b$ (gradient–intercept) or $Ax + By + C = 0$ (general)

In $y = kx + b$, k is the gradient and b the height at which the line crosses the y -axis.



The gradient is rise $\div$ run; b is where the line meets the y -axis.

A point lies on a line exactly when its coordinates **satisfy** the equation. Does $(3; 7)$ lie on $y = 2x + 1$? $2 \cdot 3 + 1 = 7$ ✓ — yes.

Through two points

Two points fix a line completely. Find the gradient first, then the intercept:

$$k = \frac{y_2 - y_1}{x_2 - x_1}$$

Through $(2; 3)$ and $(6; 11)$: $k = \frac{8}{4} = 2$, so $y = 2x + b$. Substituting $(2; 3)$ gives $3 = 4 + b$, so $b = -1$ and the line is $y = 2x - 1$. Check with the second point: $12 - 1 = 11$ ✓.

TWO SPECIAL CASES

A **horizontal** line is $y = c$ — gradient 0 . A **vertical** line is $x = c$ — it has no gradient at all, and it cannot be written as $y = kx + b$, because the run is zero.

02 Worked examples**EXAMPLE 1** Find the equation of the line through $A(1; 4)$ and $B(5; 12)$.

$$\textcircled{1} \quad k = \frac{12 - 4}{5 - 1} = 2$$

Same order top and bottom.

$$\textcircled{2} \quad y = 2x + b$$

$$\textcircled{3} \quad 4 = 2 \cdot 1 + b$$

Substitute A .

$$\textcircled{4} \quad b = 2$$

$$\textcircled{5} \quad y = 2x + 2$$

Check B : $10 + 2 = 12$ ✓**ANSWER**

$$y = 2x + 2$$

EXAMPLE 2 Where do $y = 3x - 2$ and $y = x + 6$ meet?

① $3x - 2 = x + 6$

At the meeting point both give the same y .

② $2x = 8, x = 4$

③ $y = 4 + 6 = 10$

Substitute into either equation.

④ $(4; 10)$

Check in the other: $12 - 2 = 10$ ✓

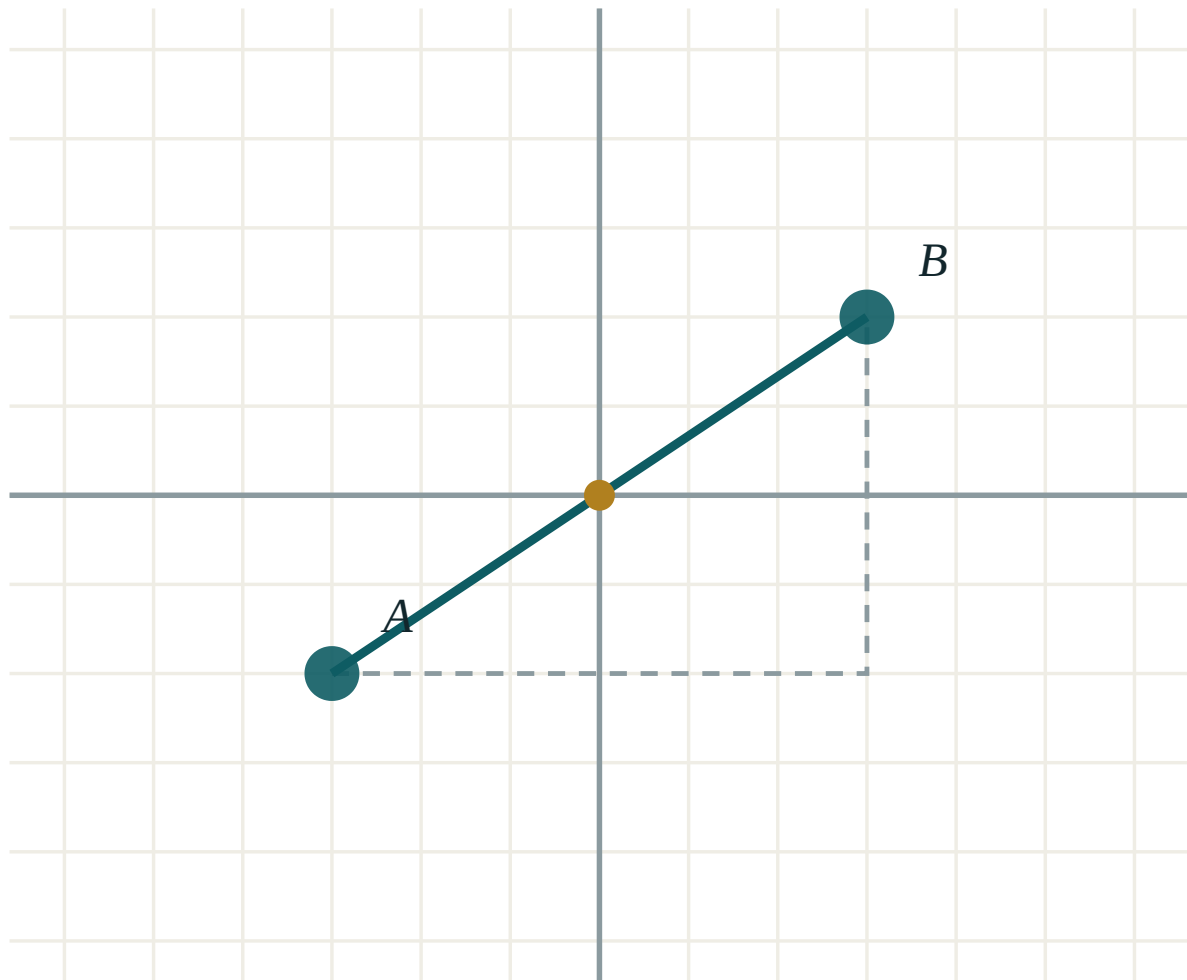
ANSWER

$(4; 10)$

03 Interactive model

Drag the two points and read the gradient from the change in x and the change in y .

A line through two points



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Straight line	To'g'ri chiziq	Прямая
Equation of a line	To'g'ri chiziq tenglamasi	Уравнение прямой
Gradient	Burchak koeffitsienti	Угловой коэффициент
Intercept	Kesma (o'qdagi)	Отрезок на оси
General form	Umumiy ko'rinish	Общий вид
Horizontal line	Gorizontal chiziq	Горизонтальная прямая
Vertical line	Vertikal chiziq	Вертикальная прямая
Satisfies the equation	Tenglamani qanoatlantiradi	Удовлетворяет уравнению
Intersection point	Kesishish nuqtasi	Точка пересечения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The gradient of $y = 4 - 2x$ is:

The line through $(0; 1)$ and $(3; 7)$ is:

A horizontal line has equation:

Does $(2; 5)$ lie on $y = 3x - 1$?

Yes

No

Only if $x > 0$

Cannot tell

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. State the gradient and intercept of $y = 3x + 5$.

ANSWER $k = 3, b = 5$

2. Does $(1; 4)$ lie on $y = 4x$?

ANSWER Yes.

3. Write the equation of the horizontal line through $(2; 7)$.

ANSWER $y = 7$

4. Write the equation of the vertical line through $(2; 7)$.

ANSWER $x = 2$

5. Find the gradient through $(0; 0)$ and $(3; 9)$.

ANSWER 3

6. Where does $y = 2x - 6$ cut the y -axis?

ANSWER $(0; -6)$

7. Where does $y = 2x - 6$ cut the x -axis?

ANSWER $(3; 0)$

Medium · the standard question

7 PROBLEMS

1. Find the equation through $(0; 3)$ with gradient -2 .

ANSWER $y = -2x + 3$

2. Find the equation through $(1; 5)$ and $(3; 11)$.

ANSWER $y = 3x + 2$

3. Find the equation through $(-2; 1)$ and $(2; 9)$.

ANSWER $y = 2x + 5$

4. Rearrange $2x + y - 8 = 0$ into $y = kx + b$.

ANSWER $y = -2x + 8$

5. Where do $y = x + 1$ and $y = 5 - x$ meet?

ANSWER $(2; 3)$

6. Find the equation through $(4; 0)$ and $(0; 8)$.

ANSWER $y = -2x + 8$

7. Does $(-1; -5)$ lie on $y = 4x - 1$?

ANSWER Yes — $-4 - 1 = -5$

Hard · stretch and reason**7 PROBLEMS**

1. Find the equation through $(2; -3)$ and $(-4; 9)$.

ANSWER $y = -2x + 1$

2. Where do $2x + y = 7$ and $x - y = 2$ meet?

ANSWER $(3; 1)$

3. Show that $(1; 3)$, $(4; 9)$ and $(6; 13)$ are collinear.

ANSWER All satisfy $y = 2x + 1$.

4. A line has gradient 3 and passes through $(2; 1)$. Find where it cuts each axis.

ANSWER $y = 3x - 5$: $(0; -5)$ and $(\frac{5}{3}; 0)$

5. Find the equation of the line through $(0; 0)$ and the midpoint of $(2; 4)$ and $(6; 8)$.

ANSWER midpoint $(4; 6)$, so $y = \frac{3}{2}x$

6. The line $y = kx + 2$ passes through $(4; -6)$. Find k .

ANSWER $k = -2$

7. Find the equation of the perpendicular bisector of the segment from $(0; 0)$ to $(4; 2)$.

ANSWER midpoint $(2; 1)$, gradient -2 : $y = -2x + 5$

07 Homework

Homework — 5 problems

Geometry 8, Tema 34, pp. 75–77. Always check your equation with the second point.

- Find the equation through $(0; -2)$ with gradient 5.
- Find the equation through $(1; 2)$ and $(4; 11)$.
- Where do $y = 3x - 4$ and $y = x + 2$ meet?
- Write the horizontal and the vertical line through $(-3; 6)$.

5. Show that $(2; 5)$, $(4; 9)$ and $(7; 15)$ are collinear.